

Module 2: Network + Network Security , Maintenance and Troubleshooting

Day 5 Troubleshooting Network Services

Task 2:

How to identify and fix it:

- Run ipconfig on the client — compare the listed **Default Gateway** against your router's actual interface IP.
- Run ping <default gateway IP> — if it fails ("Request timed out" or "Destination host unreachable"), the gateway address itself is unreachable, which points to a misconfiguration rather than a downstream routing issue.
- **Fix:** go back to the DHCP server's pool configuration, correct the Default Gateway field to match the router's real interface IP, save, then have the client release/renew its lease (ipconfig /release then ipconfig /renew, or toggle DHCP off/on in Packet Tracer) to pick up the corrected setting.

Task 3:

Names Fail, IPs Work — DNS Diagnosis

This is the classic "DNS is broken but the network itself is fine" scenario. Diagnosis steps:

1. **Confirm the split:** ping 8.8.8.8 (works) vs ping www.google.com (fails) — confirms it's specifically name resolution failing, not connectivity.
2. **Check what DNS server is configured:** ipconfig /all (Windows) — look at the DNS Servers field. If it's blank, 0.0.0.0, or an unreachable address, that's your problem.
3. **Query the DNS server directly:** nslookup www.google.com — if this hangs or returns Request timed out or No response from server, the configured DNS server isn't answering.
4. **Try a different, known-good DNS server:** nslookup www.google.com 8.8.8.8 — if this succeeds, your original DNS server (not your network) is the problem.
5. **Solution:** Update the DNS server address in your DHCP scope (or the client's static config) to a working resolver, then run ipconfig /flushdns and ipconfig /renew on affected clients.

Task 4:

Device Shows APIPA (169.x.x.x) Address

1. **Verify physical connectivity** - Confirm the physical/logical link is up — check the cable connection in Packet Tracer or the interface status light. A device with no link to the switch never even sends a DHCP Discover, so it self-assigns an APIPA address as a fallback.
2. **Test basic connectivity to the gateway** - Run ping 192.168.1.1 (or your gateway) from the affected device. If it fails immediately with no timeout delay, the interface may be down or on the wrong VLAN/subnet.
3. **Check for DHCP scope exhaustion** - Log into the DHCP server and check the pool's current lease count against Max Users. If every address is already leased out, new clients can't get one and default to APIPA. Expand the pool range or reduce lease duration if this is the cause.
4. **Verify the DHCP server is running** - Confirm the DHCP service itself is actually running (Services > DHCP > Service: On in Packet Tracer, or systemctl status isc-dhcp-server on Linux). A stopped service means no device on the network gets an offer at all.
5. **Release and renew the IP configuration** - Run ipconfig /release followed by ipconfig /renew (or toggle DHCP off then on in Packet Tracer's IP Configuration window) to force the client to send a fresh Discover message after the underlying issue is fixed.
6. **Check for a missing DHCP relay across subnets** - If the client and DHCP server sit on different subnets/VLANs, confirm an IP helper-address (DHCP relay) is configured on the intermediate router — otherwise DHCP broadcasts never reach the server across subnet boundaries.

Task 5:

Issue	Symptoms	Possible Cause	Recommended Solution
DNS — Server not responding	nslookup returns "Request timed out" or "No response from server"	DNS service crashed, wrong IP configured, or firewall blocking port 53	Restart DNS service; verify correct DNS server IP in client config; check firewall rules for UDP/TCP 53
DNS — Cached stale record	Old/wrong IP returned even after a DNS record was updated	Local or resolver DNS cache holding an outdated entry (TTL not yet expired)	Run ipconfig / flushdns (client) or clear the resolver's cache; verify the record's TTL isn't set too high
DNS — NXDOMAIN error	nslookup returns "Non-existent domain"	Typo in the domain name, or the zone/record was never created on the authoritative server	Double-check spelling; confirm the A/CNAME record exists in the correct zone file; check zone was loaded (named-checkzone)
DHCP — Client gets APIPA (169.x.x.x)	Device has no real IP, can't reach network resources	DHCP server down, scope exhausted, or no relay across subnets	Verify DHCP service is running; check pool availability; configure IP helper-address for cross-subnet requests
DHCP — IP address conflict	"Duplicate IP address detected" warning on client	Two devices assigned/statically set to the same IP, often from an overlapping static range and DHCP pool	Exclude statically-assigned IPs from the DHCP pool; use arp -a to identify the conflicting device

Issue	Symptoms	Possible Cause	Recommended Solution
DHCP – Wrong gateway/DNS handed out	Client gets an IP but can't browse the internet or reach other subnets	Misconfigured DHCP scope options (gateway or DNS server field set incorrectly)	Correct the scope options on the DHCP server; have clients release/renew (ipconfig / release then / renew) to pick up the fix